

```

int led=3;
int trig=5;
int echo=6;
long duration;
float distance;

float myMap(float input_val,float min_in,float max_in,float min_out,float
max_out)
{
    return ((max_out/max_in)*input_val);
}

void setup()
{
    pinMode(led, OUTPUT);
    pinMode(echo, INPUT);
    pinMode(trig, OUTPUT);
    Serial.begin(9600);
}

void loop()
{
    digitalWrite(trig, LOW);
    delayMicroseconds(2);

    digitalWrite(trig, HIGH);
    delayMicroseconds(10);
    digitalWrite(trig, LOW);

    duration=pulseIn(echo, HIGH);
    distance= duration*0.0343/2;
    delay(500);
    if (distance>=20 && distance<=200)
    {
        float brightness= myMap(distance, 20, 200, 0, 255);
        analogWrite(led, brightness);}
    else
    {
        analogWrite(led, LOW);}
}

```